

Adult Income Prediction — Day 1 Summary

Problem framing, hold-out test, and baseline models — UCI Adult (Census Income) dataset

1. Problem Framing

Target: income > \$50K/year = positive class. **Business objective:** identify likely high-income individuals to target for outreach (e.g. a premium product or service) while minimizing wasted contacts. **Base rate:** 23.9% positive / 76.1% negative — a moderately imbalanced problem, so raw accuracy is not a reliable success measure.

Primary metric: Precision. False positives (contacting someone who is not actually a high earner) waste outreach budget and sales effort, and with far more negatives than positives a weak model can rack up many of them. We accept lower recall (missing some real prospects) in exchange for outreach that mostly lands on genuine high-income leads. *Stakeholder framing: about 1 in 4 people in our data are high earners — we'll judge success by how many of the people we choose to contact turn out to actually be high earners.*

2. Reproducible Splits

Stratified 80/10/10 train/dev/test split, fixed random_state=42. The test set (9,769 rows) is held out and untouched until final evaluation. Reusing the test set during tuning would let the model implicitly fit its quirks, making the reported score an overly optimistic estimate of real-world performance.

3. Baseline Results (evaluated on the held-out test set)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class	0.761	0.000	0.000	0.000	0.500	0.239
capital-gain > 0 (rule)	0.781	0.617–0.625	0.22–0.225	0.33	0.59	0.32–0.33

Winner: the rule baseline. It matches the majority baseline's accuracy (~0.78 vs ~0.76) but, unlike the majority baseline, it actually identifies positives (precision ~0.62, F1 ~0.33, ROC AUC ~0.59) — the majority baseline scores a deceptively high 0.76 accuracy while being completely useless (it never predicts the positive class, ROC AUC = 0.50). A real model should clearly beat precision > 0.62–0.75 while also lifting recall and PR AUC well above the rule's 0.22 / 0.33 — otherwise it adds no value over a single feature threshold.

4. Error Analysis

False negatives (missed high earners) are dominated by people with mid-to-high education (education-num ~9–16), ages 30s–50s, and 35–55+ hours/week — legitimate high earners the rule misses entirely because their income comes from wages, not capital gains.

False positives cluster around small-to-moderate capital gains (~\$500–\$5,000) that don't actually correspond to >50K income. This exposes the rule's core weakness: it looks at exactly one feature and ignores everything else.

Fixes for tomorrow: (1) proper imputation for missing workclass/occupation/native-country instead of leaving NaN; (2) encode categoricals (workclass, occupation, marital-status, relationship) instead of discarding them; (3) address class imbalance via class weighting or resampling; (4) incorporate multi-feature signals — education, hours-per-week, occupation, relationship — instead of a single threshold; (5) handle skewed numerics like capital-gain/capital-loss (heavily zero-inflated, long-tailed).

Primary metric for the rest of the week: precision, with recall and PR AUC tracked as secondary metrics. The rule baseline already shows precision can be pushed well above a naive guess (~0.62), but its weak recall (~0.22) and PR AUC (~0.33) come from ignoring every signal except one feature. A model that raises precision above ~0.75 while also improving recall/PR AUC over the rule baseline would represent real, business-relevant progress on a 23.9% base-rate problem.